

Quantum Field Theory in Curved Spacetime

*Study Notes on Divergences, Functional Methods,
and Zeta Function Regularisation*

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Introduction

These notes document my personal study of quantum field theory in curved spacetime, following primarily the treatment in [6]. They are not a substitute for the textbook but rather a record of the calculations I worked through, the points where I got stuck, and the understanding I eventually reached.

The central theme running through these notes is the problem of *ultraviolet divergences* in quantum field theories formulated on curved backgrounds. In flat spacetime, divergences are handled by standard renormalisation; in curved spacetime, new divergences appear that are sensitive to the local geometry, and their treatment requires more sophisticated tools. The natural language for this is the *heat kernel*, a Green's function for the heat equation on the Manifold, whose short-time asymptotic expansion encodes precisely the geometric divergences of the theory. The connection between heat kernels and spectral zeta functions then provides a clean and mathematically rigorous regularisation scheme.

This connects directly to the mathematics notes on this site, where the heat kernel and its asymptotic expansion are developed from the differential-geometric side. The present notes approach the same object from the physics side: as a tool for computing one-loop effective actions and understanding the renormalisation of quantum fields in gravitational backgrounds.

Conventions. Throughout these notes we use natural units $\hbar = c = 1$. The spacetime metric has signature $(-, +, +, +)$. Einstein's summation convention is always in force. The Riemann tensor is defined by

$$[\nabla_\mu, \nabla_\nu]V^\rho = R^\rho{}_{\sigma\mu\nu}V^\sigma,$$

and the Ricci tensor by $R_{\mu\nu} = R^\rho{}_{\mu\rho\nu}$.

A note on updates. These notes are updated regularly as my studies progress. The current version covers:

- Divergences in curved spacetime
- Functional determinants and one-loop effective actions
- Zeta function regularisation

Forthcoming sections will treat the heat kernel expansion in detail, the effective potential, and applications to Yang-Mills theory.

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I Divergences in Curved Spacetime

I.1 Introduction

Quantum field theories in Minkowski spacetime generically exhibit ultraviolet divergences. In ϕ^4 theory for example, the vacuum energy density acquires a quartic divergence $\sim \Lambda^4$, the mass parameter receives a quadratic divergence $\sim \Lambda^2$, and the self-coupling is renormalised by a logarithmic divergence $\sim \log \Lambda$, where Λ is the UV cutoff.

Since spacetime is locally Minkowski and UV divergences arise from short-distance behaviour, they are governed by the local structure of the theory. One therefore expects every divergence present in flat spacetime to persist in the curved case. Beyond this, the curvature tensor introduces a new dimensionful scale into the theory, and divergences proportional to powers of curvature invariants are expected to appear. For instance, in ϕ^4 theory the scalar field has mass dimension $[\phi] = 1$ in four dimensions, and the mass counterterm $\sim \Lambda^2 \phi^2$ has dimension four. The Ricci scalar has dimension $[R] = 2$ (inverse length squared), so the combination $R\Lambda^2$ has the correct dimension four to appear as a new counterterm in the curved-spacetime theory, with no flat-space analogue. This is the first sign that renormalisation in curved spacetime requires extending the gravitational action beyond the Einstein–Hilbert term.

I.2 Euclidean Functional Integral

Consider a real scalar field ϕ propagating on a Manifold M with a fixed external metric $g_{\mu\nu}$. The action of the field is given by

$$S(\phi, g) = -\frac{1}{2} \int dx^d \sqrt{|g|} g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi. \quad (1)$$

If we consider a source j coupled to the field, then the functional integral is

$$\mathcal{Z} = \int \mathcal{D}\phi e^{i(S(\phi, g) + \int dx^d \sqrt{|g|} \phi j)} \quad (2)$$

Physical Remark. The path integral defined in equation 2 is not well defined in Minkowski signature. To see this consider the case without a source term. Field configurations with large and small kinetic energy, (action), contribute equally in amplitude to the integral, hence the oscillating behavior is intense and the integral will generically won't converge.

To ensure that the path integral formulation is better defined, One performs a Wick rotation. Consider that the metric is of the form

Derivation I.1 (Eucliden Signature).

$$g_{\mu\nu} = \begin{pmatrix} g_{00} & 0 \\ 0 & g_{ij} \end{pmatrix}, \quad (3)$$

with $g_{00} < 0$. Let $t = it_E$, then the action of the theory is

$$iS(\phi, g) = i \frac{1}{2} \int d^{d-1}x dt \sqrt{|g|} \left[-g^{00} (\partial_t \phi)^2 - g^{ij} \partial_i \phi \partial_j \phi \right], \quad (4a)$$

$$= \frac{1}{2} \int d^{d-1}x dt_E \sqrt{|g|} \left[g^{00} (\partial_{t_E} \phi)^2 - g^{ij} \partial_i \phi \partial_j \phi \right], \quad (4b)$$

$$= -S_E(\phi, g_E). \quad (4c)$$

$S_E(\phi, g)$ is the action with a Euclidean signature,

$$S_E(\phi, g) = \frac{1}{2} \int d^d x_E \sqrt{g_E} g_E^{\mu\nu} \partial_\mu \phi \partial_\nu \phi. \quad (5)$$

g_E is a positive definite metric defined on an analytically continued Manifolds M_E .

$$(g_E)_{\mu\nu} \begin{pmatrix} -g_{00} & 0 \\ 0 & g_{ij} \end{pmatrix} \quad (6)$$

Similarly we get for the source term

$$i \int d^d x \sqrt{|g|} j \phi = \int d^d x_E \sqrt{g_E} j \phi \quad (7)$$

Henceforward we drop the subscript E , unless explicitly indicated, we work in Euclidean signature.

Physical Remark. Working in the Euclidean signature does not entirely solve the problem of convergence of the action. However now field configuration with higher energy contribute smaller to the functional integral. Hence, in Euclidean signature there is better chance that the integral is well defined.

If the Manifold M is compact without boundary, than when performing integration by parts the boundary terms cancel.

Derivation 1.2 (Simplified action and Laplacian operator).

$$S(\phi, g) = \frac{1}{2} \int d^d x \sqrt{g} g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi, \quad (8a)$$

$$= \frac{1}{2} \int d^d x \sqrt{g} g^{\mu\nu} \partial_\mu (\phi \partial_\nu \phi) - \frac{1}{2} \int d^d x \sqrt{g} g^{\mu\nu} \phi \partial_\mu \partial_\nu \phi, \quad (8b)$$

$$= -\frac{1}{2} \int d^d x \sqrt{g} g^{\mu\nu} \phi \partial_\mu \partial_\nu \phi. \quad (8c)$$

The first term in equation 8b is a boundary term, hence it is equal to zero. The action can then be expressed as

$$S(\phi, g) = \frac{1}{2} \int d^d x \sqrt{g} \phi \Delta \phi. \quad (9)$$

With $\Delta = -\nabla_\mu \nabla^\mu$ is the covariant Laplacian operator.

In Euclidean Signature the functional integral is given by

$$\mathcal{Z}(j, g) = \int \mathcal{D}\phi e^{-S(\phi, g) + \int d^d x \sqrt{g} j \phi}.$$

Let $\mathcal{W}(j, g)$ be the functional defined by

$$\mathcal{Z}(j, g) = e^{\mathcal{W}(j, g)}.$$

$\mathcal{W}(j, g)$ is called the generating functional of connected Greens functions.

1.3 Functional Determinant

In this subsection we set the source to $j = 0$. In curved space time the family of exponential functions $\{e^{ik_\mu x^\mu}\}$ are no longer eigenfunction of the Laplacian operator. Instead the spectral decomposition of the operator is used. When M is compact without boundary the spectrum of the operator Δ is discrete. Let $\{\varphi_n\}_{n \in \mathbb{N}}$ be the eigenfunction of Δ ,

$$\Delta \varphi_n = \lambda_n \varphi_n, \quad (10)$$

with $\{\lambda_n\}_{n \in \mathbb{N}}$ is the corresponding family of eigenvalues.

Physical Remark. The operator $\Delta = -\nabla^2$ has a zero eigenvalue corresponding to the eigenfunction $\varphi_n = \text{const}$. This leads to an indumped term in the functional integral leading to divergences. Hence this case needs to be treated separately.

The eigenfunctions $\{\varphi_n\}_{n \in \mathbb{N}}$ are taking to be dimensionless and orthonormal with respect to the inner product on $C^\infty(M)$.

$$\langle \varphi_n | \varphi_m \rangle = \mu^d \int_M d^d x \sqrt{g} \varphi_n(x) \varphi_m(x) \quad (11a)$$

$$= \delta_{nm} \quad (11b)$$

with μ is a constant with dimension of mass.

The eigenfunctions of the operator Δ form a basis on the space of function on M . Hence the field ϕ can be decomposed as

$$\phi(x) = \sum_{n \in \mathbb{N}} a_n \varphi_n(x), \quad (12)$$

a_n are constant with the same mass dimension as ϕ . The Euclidean action is then

$$S(\phi, g) = \frac{1}{2\mu^d} \sum_{n \in \mathbb{N}} \lambda_n a_n^2 \quad (13a)$$

$$= \frac{1}{2} \sum_{n \in \mathbb{N}} \tilde{\lambda}_n \tilde{a}_n^2 \quad (13b)$$

with

$$\tilde{\lambda}_n = \frac{\lambda_n}{\mu^2} \quad , \quad \tilde{a}_n = \frac{a_n}{\mu^{\frac{d}{2}-1}}.$$

The path integral measure is then $\mathcal{D}\phi = \mathcal{N} \prod_{n \in \mathbb{N}} \frac{d a_n}{\mu^{\frac{d}{2}-1}} = \mathcal{N} \prod_{n \in \mathbb{N}} d \tilde{a}_n$. $\mathcal{N}$ is a renormalization constant such that

$$\int \mathcal{D}\phi \exp \left[\frac{-\mu^2}{2} \int d^d x \sqrt{g} \phi^2 \right] = 1 \quad (14)$$

Derivation 1.3 (Formal expression of $\mathcal{N}$). Consider first the integral

$$\int d^d x \sqrt{g} \phi^2 = \mu^{-d} \sum_{n,m \in \mathbb{N}} a_n a_m \mu^d \int d^d x \sqrt{g} \varphi_n \varphi_m, \quad (15a)$$

$$= \mu^{-d} \sum_{n,m \in \mathbb{N}} a_n a_m \langle \varphi_n | \varphi_m \rangle, \quad (15b)$$

$$= \mu^{-2} \sum_{n \in \mathbb{N}} \tilde{a}_n^2. \quad (15c)$$

Injecting this result in equations 14, we get

$$1 = \int \mathcal{D}\phi \exp \left[\frac{-\mu^2}{2} \int d^d x \sqrt{g} \phi^2 \right], \quad (16a)$$

$$= \mathcal{N} \prod_{n \in \mathbb{N}} \int d \tilde{a}_n \exp \left[-\frac{1}{2} \sum_{n \in \mathbb{N}} \tilde{a}_n^2 \right], \quad (16b)$$

$$= \mathcal{N} \prod_{n \in \mathbb{N}} \left[\int d \tilde{a}_n e^{-\frac{1}{2} \tilde{a}_n^2} \right], \quad (16c)$$

$$= \mathcal{N} \prod_{n \in \mathbb{N}} \sqrt{2\pi}. \quad (16d)$$

In going from 16c to 16d we used the Gaussian integral formula. Equation 16d implicitly defines the value of the normalization constant $\mathcal{N}$.

The path integral with zero source can then be expressed as a formal determinant of the scaled operator $\tilde{\Delta} = \frac{1}{\mu^2} \Delta$.

$$\mathcal{Z}(0, g) = e^{\mathcal{W}(0, g)} = \mathcal{N} \prod_{n \in \mathbb{N}} \left[\int d \tilde{a}_n e^{-\frac{1}{2} \tilde{\lambda}_n \tilde{a}_n^2} \right], \quad (17a)$$

$$= \mathcal{N} \prod_{n \in \mathbb{N}} \sqrt{\frac{2\pi}{\tilde{\lambda}_n}} \quad (17b)$$

$$= \det(\tilde{\Delta})^{-\frac{1}{2}} \quad (17c)$$

2 Heat Kernel Expansion

Remark. This section is in preparation. It will cover the Seeley–DeWitt asymptotic expansion

$$\mathrm{Tr} \, e^{-tA} \sim (4\pi t)^{-n/2} \sum_{k=0}^{\infty} a_k(A) t^k \quad \text{as } t \rightarrow 0^+,$$

the computation of the first few coefficients a_0, a_1, a_2 in terms of geometric invariants, and their role as counterterms in curved spacetime QFT. See the mathematics notes for the purely geometric treatment.

3 Effective Potential

Remark. This section is in preparation. It will derive the one-loop effective potential from the zeta-regularised determinant, discussing its renormalisation and the Coleman-Weinberg mechanism in curved spacetime.

References

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